LEGEND

Visualizing Aggregations in STAC

From scenes to coverage

Matt Hanson

FOSS4G North America 2026 · Sacramento

Nov 3, 2026

Do I have data?

Do I have data?

- Sentinel-2
- over this county
- last summer
- under 20% cloud

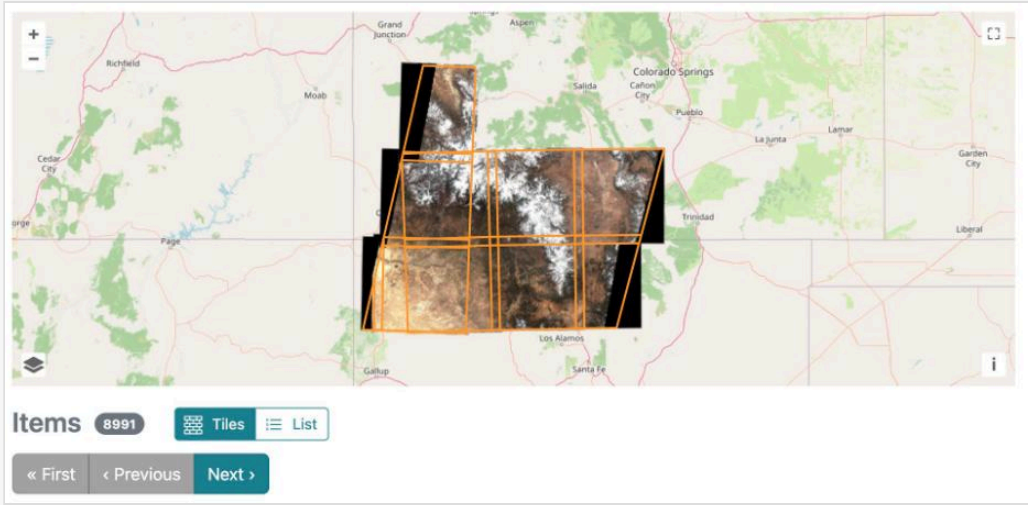
Probably yes. But how much, when, and is any of it usable?

STAC gives you items

```
GET /search
  ?collections=sentinel-2-c1-l2a
  &bbox=-109.06,36.99,-102.04,41.0
  &datetime=2025-01-01/2025-12-31
```

- one item per scene
- 100 per page
- follow `next`

Colorado, 2025: **8,991** items



I don’t want 8,991 items

I want to know

- where
- when
- how many

01 The Aggregation extension

Two endpoints

`GET /aggregations`	what you can ask for, like `/queryables`
`GET /aggregate`	the `/search` parameters, plus `aggregations=`
`/collections/{id}/aggregations` `/collections/{id}/aggregate`	the same, for one collection

```
{ "aggregations": [
  { "name": "total_count",
    "data_type": "integer" },
  { "name": "datetime_frequency",
    "data_type": "frequency_distribution" },
  … ] }
```

- found by `rel=aggregate`, `rel=aggregations` links
- conformance `…/v0.3.0/aggregation`
- back: a `value`, or `buckets` of `key` + `frequency`

Ask for the shape instead

```
GET /aggregate?collections=sentinel-2-c1-l2a&bbox=-109.06,36.99,-102.04,41.0&datetime=...
    &aggregations=total_count,datetime_frequency
```

Same parameters as `/search`. Buckets back, one response.

```
{ "aggregations": [
  { "name": "total_count", "value": 8991 },
  { "name": "datetime_frequency", "buckets": [
      { "key": "2025-01-01T00:00:00.000Z", "frequency": 594 },
      { "key": "2025-02-01T00:00:00.000Z", "frequency": 595 }, … ] } ] }
```

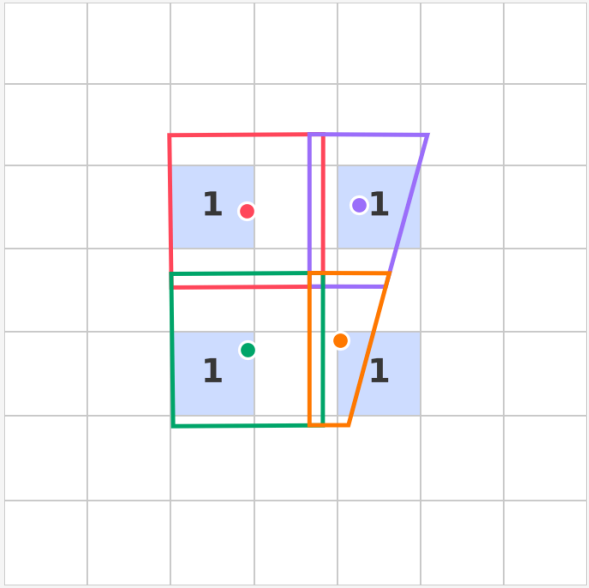
What you can ask for

	spec	stac-server adds
how many	`total_count`	
when	`datetime_min` `datetime_max` `datetime_frequency`	
what	`collection_frequency` `cloud_cover_frequency`	`platform_frequency` `sun_elevation_frequency` `grid_code_frequency` …
where		`centroid_{geohex,geotile,geohash}_grid_frequency` `geometry_{geotile,geohash}_grid_frequency`

I think *where* is the useful part. None of it is in the spec.

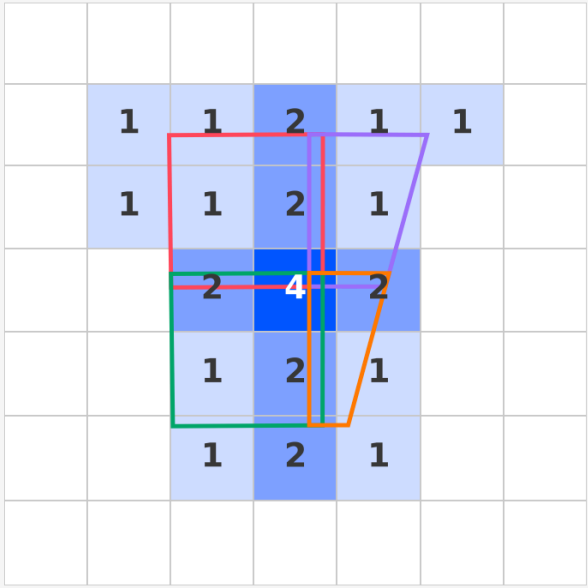
Where: centroid or footprint

centroid: once, where its center falls



4 scenes → counts sum to 4

geometry: every cell it touches

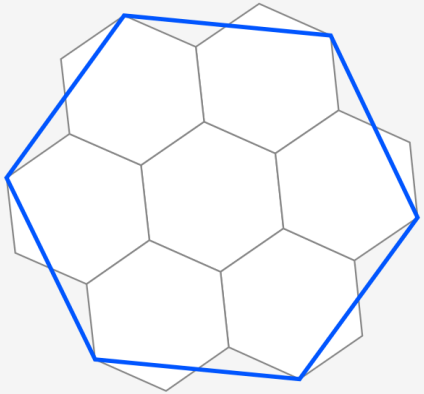


4 scenes → counts sum to 27

Three grids

H3 (geohex)

parent `83268cfffffffff`



hexagons, about equal area
each splits into 7
resolution 0–15

map tiles (geotile)

parent `8/53/97`

9/106/194	9/107/194
9/106/195	9/107/195

web-map squares (Mercator)
each splits into 4
zoom 0–29

geohash

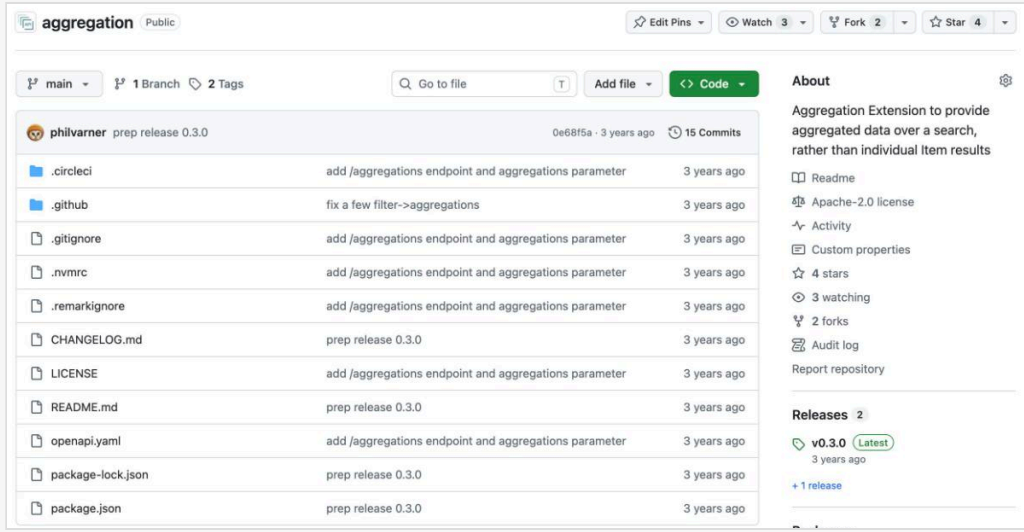
parent `9xj`

9xjp	9xjr	9xjx	9xjz
9xjn	9xjq	9xjw	9xjy
9xjj	9xjm	9xjt	9xjv
9xjh	9xjk	9xjs	9xju
9xj5	9xj7	9xje	9xjg
9xj4	9xj6	9xjd	9xjf
9xj1	9xj3	9xj9	9xjc
9xj0	9xj2	9xj8	9xjb

lat/lon rectangles, named by a string
each extra letter splits into 32
length 1–12

Where the spec is

- v0.3.0, still a **Proposal**
- last change April 2023
- no OpenAPI
- stac-server: GET only



02 # stui

Counts first

- **Search**: aggregations only
  - total, monthly histogram, density grid
  - fast at any size
- **Get footprints**: a separate button
  - all of them, or a month at a time

`EarthLegend/stui` · MapLibre · an embeddable React component · Apache-2.0

Who aggregates what

the server, on Search

- how many
- by month
- where: H3 hexes, map tiles
- every match counts

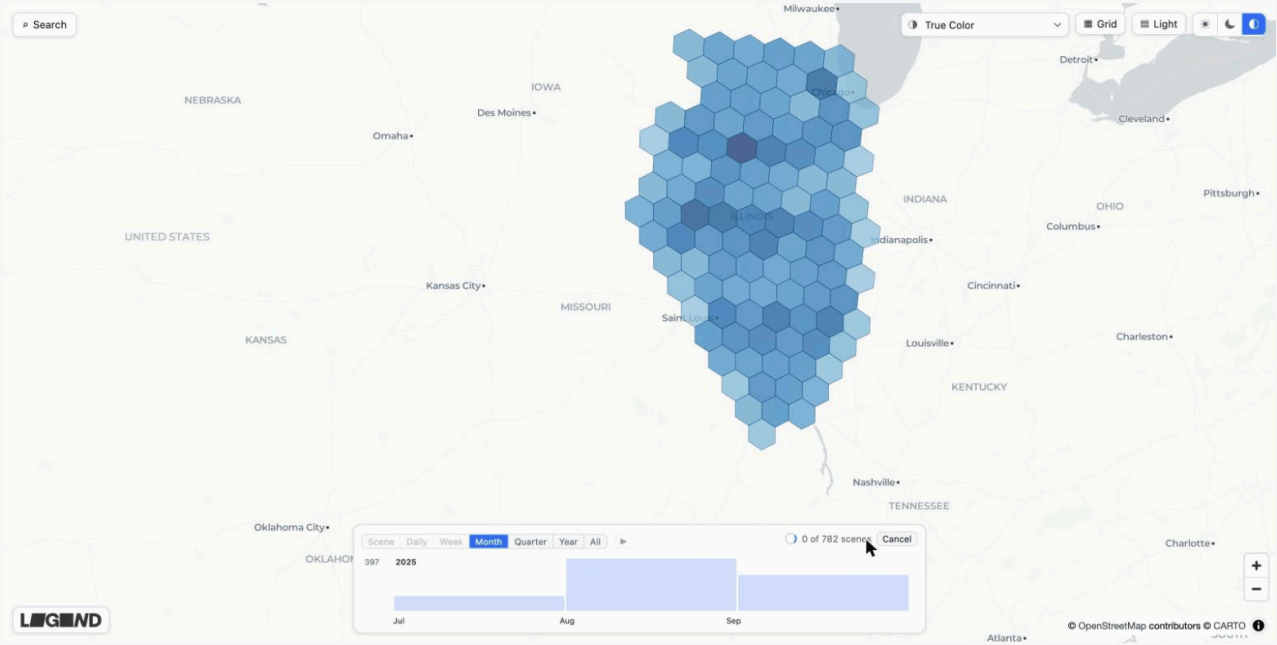
the browser, once footprints load

- by day, by week
- solar day
- one shape per period
- only what’s loaded

Demo

- NAIP, Illinois, 2025
- hexes → map tiles → footprints
- by day, by week

Backup: the demo



In the browser

/search, geometry only → **group by solar day** → **union each group** → **MapLibre**

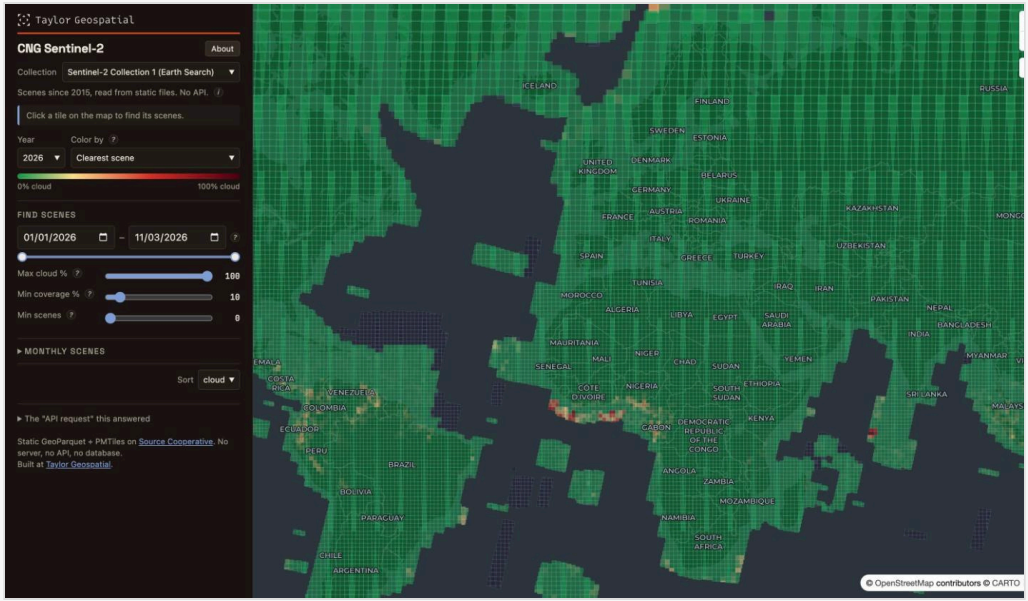
- drop footprints that add no new ground first: Sentinel-2, 554 → 26
- tens of thousands of footprints on the GPU, no bounding boxes
- solar day = UTC + longitude / 15 h

03

How does this compare to GeoParquet + DuckDB?

Chris Holmes's Sentinel-2 explorer

- stac-geoparquet on Source Cooperative, a mirror of Earth Search
- about 30 million scenes, refreshed daily
- parquet read in the browser
- no API



How it gets coverage

- precomputed: one row per MGRS tile per month
  - scene count, cloud cover, how much of the tile is covered
- click a tile: read its items for the year, filter in the browser

Same question, in DuckDB

```sql
SELECT _month, count(*)
FROM read_parquet('https://data.source.coop/tge-labs/s2-stac-geoparquet/
                   sentinel-2-c1-l2a/year=2025/items.parquet')
WHERE bbox[1] <= -102.04 AND bbox[3] >= -109.06
  AND bbox[2] <= 41.0    AND bbox[4] >= 36.99
GROUP BY 1
```

8,991. Same answer, same months.

How long it takes

	Earth Search	DuckDB + parquet
Colorado, 2025, by month	0.2 s	17 s first query, 0.8 s after
the world, 2025, by month	0.5 s	15 s
the world, 2025, H3	0.5 s	15 s
whole archive, by month	0.3 s	minutes

- the parquet time is downloading, not counting
- Colorado gets fast because the file is sorted by location

my laptop, one afternoon, 2026-11-02

Which do you want?

GeoParquet + DuckDB

- nothing to run
- any question you can write in SQL
- fast for small areas and precomputed views
- as fresh as the last build

stac-server

- about a second, cold, any area
- only the aggregations the server offers
- any collection on the server, as fresh as ingest
- a standard any client can speak

Coming in stac-server

- `datetime_frequency_interval`
  - day, week, month, quarter, year
- on `main`, not released yet

What's next for stui

- more than one collection
  - stacked histogram, colored by collection
- a table, and export
  - click a hex or a month, get the scenes
- cloud cover as an aggregation, not just a filter
- daily counts from the server

If you run a STAC API

- turn on the Aggregation extension
- give your items `proj:centroid`
- pgstac doesn't have it yet. Anyone want to?



Thank you

SLIDES

geoskeptic.dev/presentations/2026/visualizing-stac-aggregation

matt@lgnd.ai